

TrES-3b

R-2779

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130707 · created 2026-09-03T23:53:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456480.8816 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.85 +/- 2.25 [%]
Orbital Inclination (inc)	86.99 +/- 2.89
Airmass coefficient 1 (a1)	0.773 +/- 0.068
Airmass coefficient 2 (a2)	0.14 +/- 0.2
Scatter in the residuals of the lightcurve fit is	9.21 %
Best Comparison Star	#3 - [518, 180]
Optimal Aperture	2.24
Optimal Annulus	6.58
Transit Duration (day)	0.0594 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.057 vs 0.16309 (deviation 0.58σ)
Duration fit vs published	0.0594 d vs 0.0567 d (deviation 0.45σ)
Mid-time O–C	1.0 min
Depth SNR	3.4
Residual scatter	9.21 %

Light curve

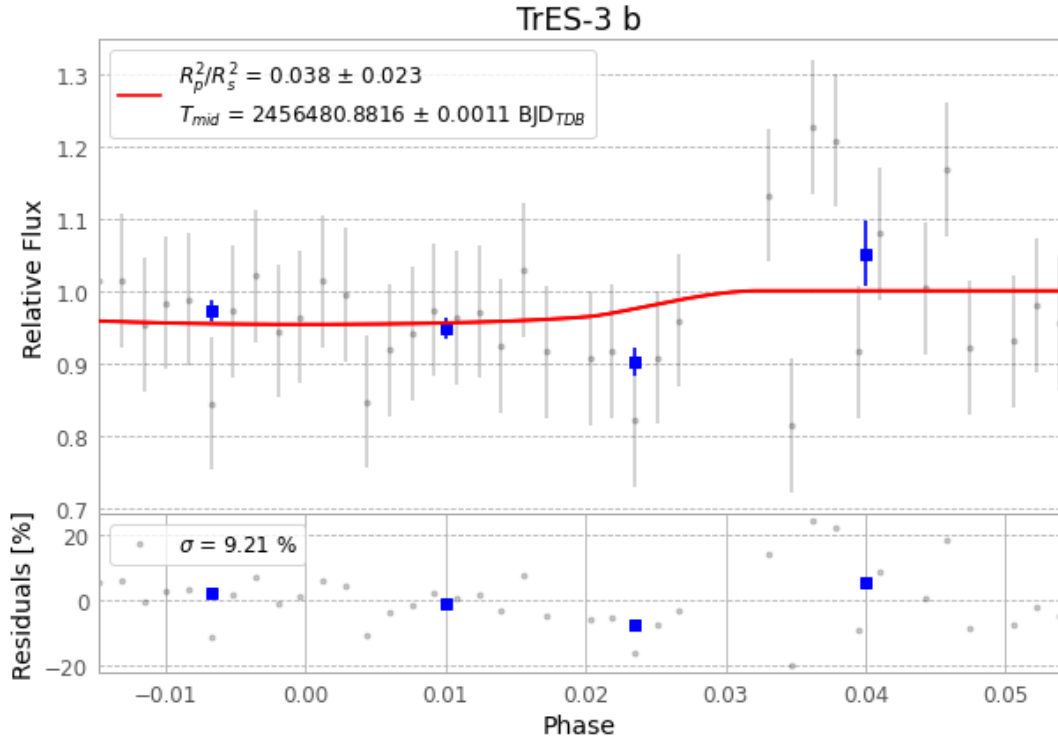


Figure 1 — FinalLightCurve_TrES-3 b_2013-07-07.png (SHA-256 40ae10424cea7ca3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 203242501b45f0f9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

44 FITS frames (29 MB), TRES-3130707083612.FITS → TRES-3130707104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130707.log (ae987ace77ad...), FinalLightCurve_TrES-3 b_2013-07-07.png (40ae10424cea...), FinalLightCurve_TrES-3 b_2013-07-07.csv (de456f06b763...)

Compute resources

Wall time 413.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:53Z.